SUZHE XU

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EDUCATION

Huaqiao University Sep. 2023 – Jun. 2027 (expected)
Undergraduate, Artificial Intelligence, School of Computer Science and Technology
Comprehensive evaluation rank (2023–2025): 1/40; academic rank: 4/40.
GPA: 4.09/5.00 (84.77/100).

ACHIEVEMENTS

National Scholarship, Ministry of Education of China, 2024
— awarded to approximately 0.3% of university students nationwide.
First Prize (Provincial Level), China Undergraduate Mathematical Contest in Modeling (CUMCM).
Completed project, National Undergraduate Innovation and Entrepreneurship Training Program (China).

RESEARCH INTERESTS

Efficient generative modeling: native few-step generation, diffusion distillation, video inverse problems, and streaming world models; training stability and multimodal post-training.

PUBLICATIONS AND MANUSCRIPTS

1. Weimin Bai*, **Suzhe Xu***, Yiwei Ren, Jinhua Hao, Ming Sun, Wenzheng Chen, and He Sun. **InstantViR: Real-Time Video Inverse Problem Solver with Distilled Diffusion Prior.** *CVPR*, 2026. **Highlight**. Co-project lead and presenter. *Equal contribution. [Paper] [Project] [Presentation]
2. **Suzhe Xu***, Weimin Bai*, Jinhua Hao, Ming Sun, Wenzheng Chen, and He Sun. **InstantViR++: Improved Posterior Distillation for Real-Time Video Inverse Problem Solving.** *Submitted to IEEE Transactions on Multimedia*, 2026. *Equal contribution.
3. **KAT-Coder-V2 Technical Report.** arXiv:2603.27703, 2026. [Report]
   Contributions: execution-based WebCoding data curation, supervised fine-tuning, and long-context continued pretraining.

RESEARCH EXPERIENCE

Peking University Jun. 2025 – Present
Visiting Student / Research Assistant, Wangxuan Institute of Computer Technology
Computational Imaging Lab; advised by He Sun and Wenzheng Chen.

InstantViR and InstantViR++
Real-time video inverse problems

- To overcome the inference cost of iterative diffusion posterior sampling, co-led the design and development of InstantViR, distilling a video diffusion prior into a causal one-step solver for deblurring, inpainting, and super-resolution.
- Developed the distillation pipeline, causal autoregressive solver, and LeanVAE integration. The published system exceeds **35 FPS**, with up to **100× speedup** over iterative solvers, without external paired clean/degraded training videos.
- Investigated teacher–student latent-space mismatch and competing supervision in InstantViR++, using bidirectional latent conversion and cyclic loss weighting to balance measurement consistency and generative supervision. Evaluated the design through controlled comparisons; manuscript submitted to **IEEE Transactions on Multimedia**.

RESEARCH EXPERIENCE (CONTINUED)

Streaming World Models for Autonomous Driving Jun. 2026 – Sep. 2026
Industry-collaborative research project Peking University

- To reduce the inference cost of streaming driving-video generation, developed a bidirectional-teacher → causal-teacher → consistency-distilled student pipeline for Waymo videos, enabling **four-step block-autoregressive generation**. Designed a clean KV cache to condition subsequent blocks on committed clean history.
- Addressed training–inference mismatch through differentiable self-forcing rollouts and **25/49-frame training**. Used ablations to diagnose exposure bias and extended the pipeline to multi-view generation.
  Prometheus: Controllable Video World Models for Autonomous Driving. Technical report in preparation; working title.

Multi-Agent World Model Post-Training and Evaluation Ongoing
Research project Peking University

- To model multiple agents under independent action inputs, adapted Cosmos3-Edge for native dual-view generation using agent-specific action tokens and four-axis positional encoding. Completed staged training with a **93-to-189-frame curriculum** on **16 A100 GPUs across two nodes**.
- Established protocol-matched comparisons with the Gamma teacher by aligning frame rates, prediction horizons, and sampling budgets. Identified persistent material and geometric errors despite lower validation loss, highlighting a gap between the training metric and visual fidelity.

Rednote (Xiaohongshu), Dots Studio Aug. 2026 – Oct. 2026
Multimodal Research Intern — Ace Top Talent Internship Program

- **Native few-step language generation:** Investigate how few-step generative objectives transfer to language modeling. Establish FLM cross-entropy/MSE baselines and adapt MeanFlow and simplified consistency training; compare **1/2/4-step generation** using Gen PPL, MAUVE, repetition rates, and samples to examine objective–quality discrepancies and quality–diversity–compute trade-offs.
- **Distribution-matching distillation:** Investigate how score-difference processing affects one-step diffusion distillation using norm-preserving spectral transformations and rotation-based adaptations. Build reproducible CIFAR-10 experiments to disentangle residual-space processing, auxiliary score-network tracking, and student optimization; evaluate sample quality, diversity, and convergence.
- Support controlled comparisons through coordinated **8-GPU experiments**, with peak aggregate usage exceeding **100 GPUs**; manage experiment tracking, checkpointing, and evaluation across runs.

ADDITIONAL INDUSTRY EXPERIENCE

Kuaishou Jan. 2026 – Jun. 2026
Algorithm Research Intern, Frontier Agent Algorithms

- To provide functionally verified training data for WebCoding, built a WebGen-Bench-based pipeline combining LLM-generated tests, browser automation, and WebVoyager UI agents. Curated **10k+ web-tool/game examples**, participated in SFT training, and extended evaluation to 3D scenarios.
- To support repository-level editing and long-context code modeling, built an Issue–PR-to-FIM/unified-diff pipeline with 64K/256K packing, producing **187,858 examples from 3,600+ repositories**. Trained and evaluated Qwen3.5-35B-A3B at 64K context using Megatron-Bridge on **32 H800 GPUs**.
  Also developed InstantViR++ during this internship with Jinhua Hao (Kling AI) and Peking University, using Kuaishou compute resources; research details appear above.

SKILLS AND LANGUAGES

Technical Skills: PyTorch, distributed GPU training, FSDP, LoRA, Megatron-Bridge, TensorBoard, Git, LaTeX.
Languages: Mandarin Chinese; English (IELTS 7.5 overall); French (B1, self-assessed).